

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 14

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 2 from:</i> • Potato bean (N America): high protein content • Morama bean (southern Africa): protein content similar to soya, drought resistant • Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils • Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion • American bison: graze grassland too poor for cattle • Common eland: browse and graze so can be kept with cattle without severe competition	2
01.2	• Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil • So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted	2
01.3	• They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
02.1	<i>Any 3 from:</i> • Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses • Setting lines at night when birds are not feeding • Bird-scaring streamers / decoys • Curved hooks that catch tuna but not birds	3

Question	Marking guidance	Marks
02.2	• Effort: limits on boat size, engine power, days at sea, number of licences; decommissioning payments to reduce fleet size where overfishing is unavoidable • Methods: bans or length limits on drift nets; bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) • Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes)	3
02.3	• Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) • Juveniles are released when survival chances are good, increasing recruitment to the adult population	2
02.4	• Egg / larval survival varies greatly between years so adult numbers do not predict recruitment • Several poor years would lower the MSY; egg surveys allow catches to be reduced in advance rather than after a sudden collapse	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
03.1	• $2.7 - 2.45 = 0.25$ billion = 250 million years	1
03.2	• For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans • Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere	2
03.3	<i>Any 3 from:</i> • Oxygen in the atmosphere absorbed ultraviolet light • A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed • Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) • Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out	3
03.4	• Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) • The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years)	2
03.5	<i>Any 2 from:</i> • Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion • Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas • Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	

Question	Marking guidance	Marks
04.1	• Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change • No single source could meet demand all year; battery storage on the island grid smooths fluctuations • Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated)	3
04.2	<i>Any 3 from:</i> • Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife • Wood biomass heating; insulation such as straw-bale housing; some electric vehicles • Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting • Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection	3
04.3	• Small groups can organise, agree and act quickly; decisions benefit people who see the results directly • Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away	2
04.4	• Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand • DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	
05.1	• Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs	1
05.2	• Impact depends on total population x per capita impact • With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added • Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates	3
05.3	• Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage • Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need	2
05.4	• Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity • In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) • Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
06.1	• B	1
06.2	<i>Any 4 from:</i> • Deforestation • Ploughing of grassland / expansion of farmland • Reservoir creation / flooding • Mineral extraction, especially open-cast mining • Urban expansion	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	
07.1	• Very small gene pool so individuals are closely related and inbreeding is likely • Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low • A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) • Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies	4
07.2	<i>Any 3 from:</i> • Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided • Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life • Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) • Stress / disturbance	3
07.3	• AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection • Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) • So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
08.1	• Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates • Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials • Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult • Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable	4

Question	Marking guidance	Marks
08.2	- Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life - They leach from landfill or are released by incineration; persistent, bioaccumulating	2
08.3	- Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) - Easy identification of materials (coding) and separation of components so they can be recycled - Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) - Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
09.1	- Upwelling currents bring nutrients from the seabed to surface waters 24-hour daylight in summer allows a huge phytoplankton bloom - Algae also grow under sea ice where krill shelter and feed; krill support most of the food web	3
09.2	- Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web - Toothfish overfishing removes the food of sperm whales and elephant seals - Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch)	3
09.3	- Ice on land stores 70% of the Earth's fresh water, keeping sea levels low - High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming) - Carbon sequestration: carbon in algae eaten by krill sinks to the seabed in krill faeces	3
09.4	Stronger winds spreading ice; more snowfall from warmer, moister air freezing onto slush; fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4.	
10.1	- About 10% of land is already cultivated and 25% grazed - Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil	2
10.2	- Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies - So people with less money lose access to the land and food they used to produce	2
10.3	- It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services - So that pesticides and artificial fertilisers become unnecessary rather than just being banned	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Unsustainable features: reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • Pest control: cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • Nutrients: recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • Water: low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • Soil and carbon: low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • Biodiversity: conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland; agri-environment schemes • Evaluation: growing population (perhaps 12 billion) needs more food; lower trophic level diets; obstacles - cost, need for support, gaining acceptance of new techniques; learning from past land degradation (fertile crescent, US mid-west) • Judgement: no single strategy sufficient; combination of ecological principles and new technology; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Definition of sustainability (Brundtland); population x per capita impact; dynamic equilibria, feedback, tipping points, diversity • Principles of natural systems: renewable low-density energy, low temperatures, material cycles, non-toxic wastes; contrast with linear human systems • Circular economy principles: cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • Measuring progress: ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • Barriers: economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	25

Question	Marking guidance	Marks
11.2	Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations; positive feedback and tipping points caused by humans • Energy: solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process); renewable energy, heat pumps, bioleaching, fuel cells • Materials: cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals; recycling, Cradle to Cradle, industrial symbiosis • Diversity and stability: rainforests / reefs vs agroecosystems; diverse energy mixes • Optimum vs maximum production; supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics; limits and judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	25